Knowledge City:

Components In The App:

- SPA (Front end)
- PHP App → Buss Logic
- Clickhouse → data & reporting
- Video Conversion
- future Microservice (to be added)

[Backend]

[Cloud Provider: AWS]

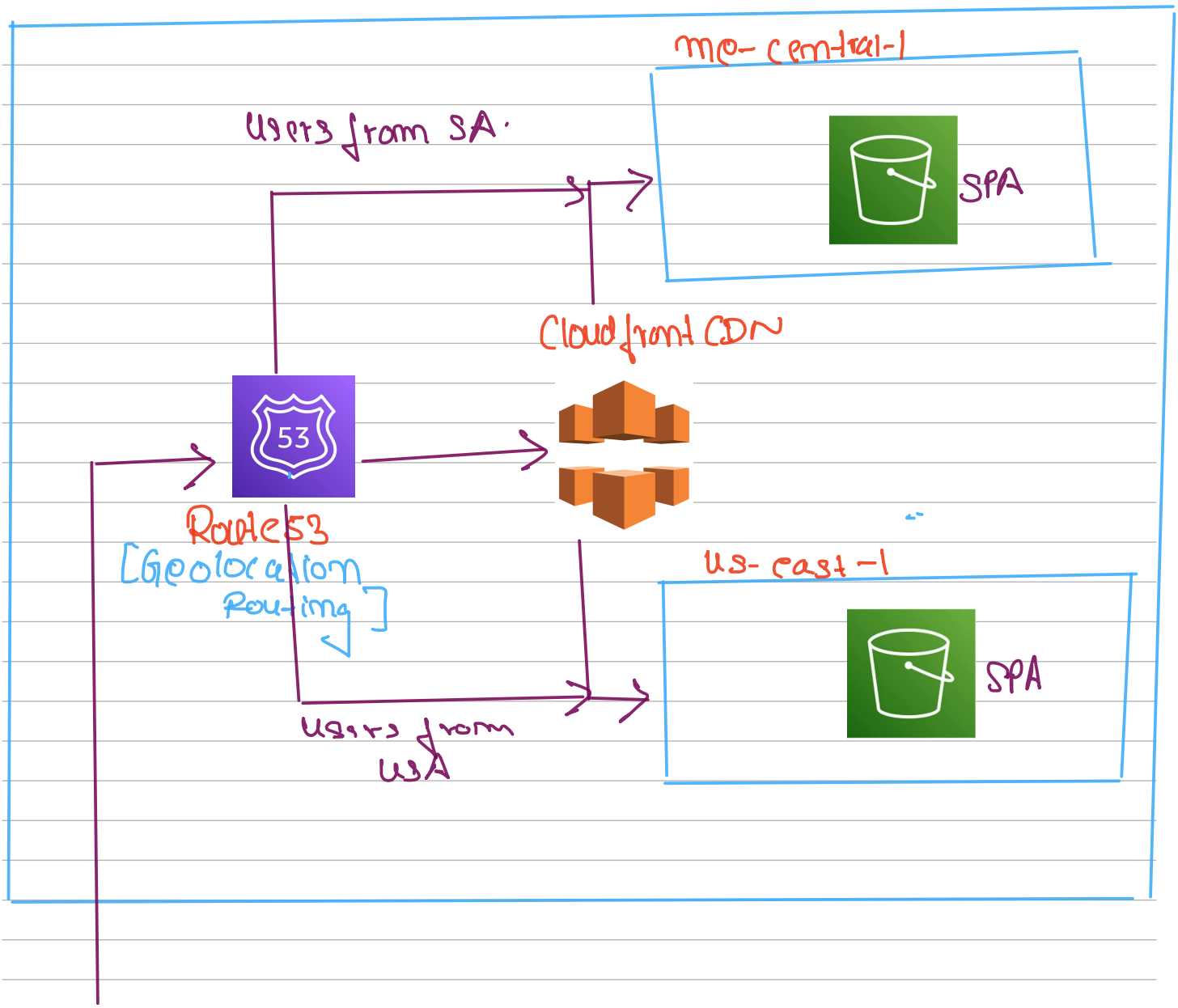
(A) Architectural Design:

Frontend Application [SPA]

↓

AWS Service - S3 bucket → static Website hosting

- Saudi Arbia Region — me-central-1
  - Multi-AZ
    - // 1a
    - // 1b
    - // 1c
- US region — us-east-1
  - // 1a
  - // 1b
  - // 1c
  - // 1d
  - // 1e
  - // 1



Why these services?

↳ CDN = Distributes content with low latency.
↳ Route53 = Geolocation Routing

Users from SA → me-central-1
" " USA → us-east-1

xxxx

for High Availbity → Multi-AZ
for DR → Multi-Region

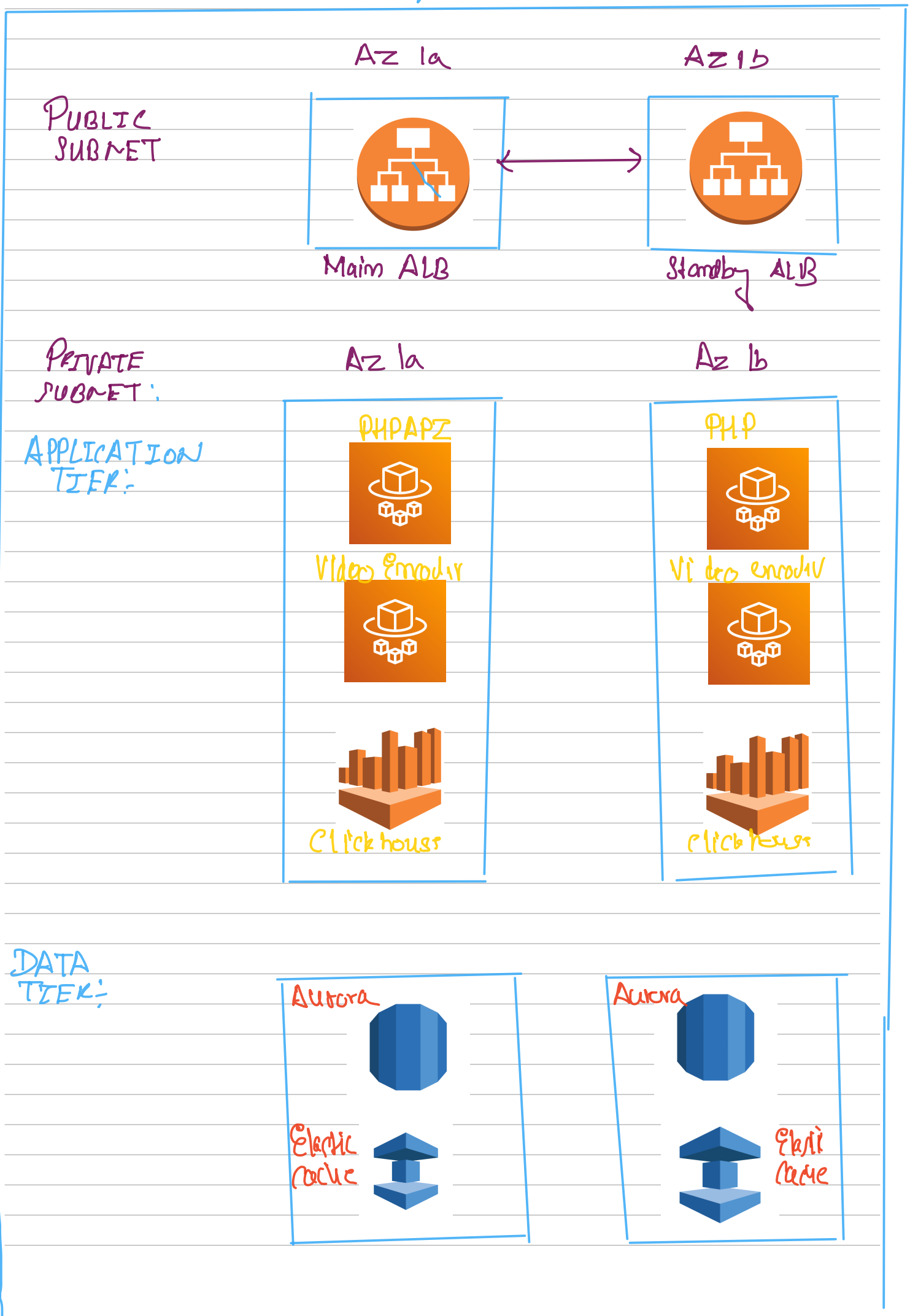
BACKEND APPLICATION:

1. PHP APP [Business Logic]
   ↓
   AWS service = Containerized → Docker
   on
   → ECS → Fargate

2. Clickhouse [Data & Reporting]
   → Analytics

3. Video Converter
   → Fargate

me-central-1 / us-east-1



STORAGE TEEK:-

S3 bucket



[User Data bucket, Raw Videos]

(B) Security & Monitoring:-



IAM



Secret Manager

Security Compliance



Config



Guard Duty.

For Monitoring:

Cloudwatch & AWS X-Ray

can be enabled at the Backend App level.